

Social Network data analysis for community detection and influential analytics for given problem using Girvan newman algorithm / Kmean clustering algorithm

```
import pandas as pd
import networkx as nx
from networkx.algorithms import community
from sklearn.cluster import KMeans
from sklearn.preprocessing import LabelEncoder
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('tweets.csv')
df.head()
```

	created_at	tweet	user	user_location	retweet_count	additional_data	
0	05-02-2023 11:37	Anyone who Belongs To The Society,\nWhoever Wi...	PVRTWEETSSS	NaN	0	{'created_at': 'Sun Feb 05 11:37:50 +0000 2023...	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	
2	05-02-2023 10:03	And people call him bad?\n#tate #FreeTheTates ...	hsenshamass	NaN	0	{'created_at': 'Sun Feb 05 10:03:48 +0000 2023...	
3	05-02-2023 08:34	How's this poetry or you can call this poem. #...	kshitiz_999	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 08:34:39 +0000 2023...	
4	05-02-2023 08:19	Andrew Tate's Best Piece of advice...\n\nUnleash...	TheWolfCeo	NaN	0	{'created_at': 'Sun Feb 05 08:19:04 +0000 2023...	

Next steps:

[Generate code with df](#)
[View recommended plots](#)
[New interactive sheet](#)

```
df = df[df.user_location.notnull()][:10]
df['hashtags'] = ''
df.head()
```

	created_at	tweet	user	user_location	retweet_count	additional_data	hashtags	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...		
3	05-02-2023 08:34	How's this poetry or you can call this poem. #...	kshitiz_999	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 08:34:39 +0000 2023...		
24	04-02-2023 22:06	@Cobratate Calm down kickboxer boy! We have bi...	after_shook	Place(_api=<tweepy.api.API object at 0x000001B...	2	{'created_at': 'Sat Feb 04 22:06:17 +0000 2023...		
31	04-02-2023	@Cobratate I'm anti	after_shook	Place(_api=<tweepy.api.API object at 0x000001B...	1	{'created_at': 'Sat Feb 04		

```
for i, row in df.iterrows():
    hashtags = [token for token in row.tweet.split() if token.startswith('#')]
    df['hashtags'][i] = hashtags
df.head()
```

Show hidden output

```
df = df.explode('hashtags')
users = list(df['user'].unique())
hashtags = list(df['hashtags'].unique())
df.head()
```

```
print('Top 5 Most Influential Users/Hashtags based on betweenness centrality :- ')
influential_nodes = sorted(
    centrality.items(), key= lambda item: item[1], reverse=True
)[:5]
print(centrality.items())
```

```
Top 5 Most Influential Users/Hashtags based on betweenness centrality :-
dict_items([('Immortalayush_', 0.1970310391363023), ('kshitiz_999', 0.3731443994601889), ('after_shook', 0.3724696356275303), ('SARS
```

```
for nodes in influential_nodes:
    print(f"{nodes[0]}: {nodes[1]}")
```

```
#AndrewTate: 0.7584345479082321
SARSALI99: 0.45276653171390013
idilaberentur: 0.45209176788124156
kshitiz_999: 0.3731443994601889
after_shook: 0.3724696356275303
```

```
label_encoder = LabelEncoder()
```

```
df['user_label'] = label_encoder.fit_transform(df['user'])
df['hashtag_label'] = label_encoder.fit_transform(df['hashtags'])
df.head()
```

	created_at	tweet	user	user_location	retweet_count	additional_data	hashtags	user_label	hashtag
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#NewProfilePic	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#AndrewTate	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#khabib	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#ConorMcGregor	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#AskSRK	0	

```
X = df[['user_label', 'hashtag_label']]
X.head()
```

	user_label	hashtag_label
1	0	10
1	0	0
1	0	21
1	0	2
1	0	1

Next steps: [Generate code with X](#) [View recommended plots](#) [New interactive sheet](#)

```
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X)
df['cluster'] = kmeans.labels_
df.head()
```

	created_at	tweet	user	user_location	retweet_count	additional_data	hashtags	user_label	hashtag
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#NewProfilePic	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#AndrewTate	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#khabib	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#ConorMcGregor	0	
1	05-02-2023 10:51	#NewProfilePic\nKeep Grinding, for the loved o...	Immortalayush_	Place(_api=<tweepy.api.API object at 0x000001B...	0	{'created_at': 'Sun Feb 05 10:51:44 +0000 2023...	#AskSRK	0	

```
plt.scatter(df['user_label'], df['hashtag_label'], c=df['cluster'], cmap='viridis')
plt.xlabel('User Labels')
print('Hashtags Labels')
plt.title('K-Means Clustering')
plt.colorbar(label='Cluster')
plt.show()
```

